

Matrix multiplication

Matrix multiplication is multiplying corresponding elements from the row of the first matrix by the elements from the column of the second matrix, adding the products together.

Learning on the example matrices, if we multiply row 3 from A by column 3 from B, we get:

$$a_{31} \cdot b_{13} + \dots + a_{3n} \cdot b_{n3}$$

For an example with full matrices, suppose we have:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}, B = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{bmatrix}_{3 \times 4}, \text{ then } AB =$$

$$\begin{bmatrix} 1a+2e+3i & 1b+2f+3j & 1c+2g+3k & 1d+2h+3l \\ 4a+5e+6i & 4b+5f+6j & 4c+5g+6k & 4d+5h+6l \\ 7a+8e+9i & 7b+8f+9j & 7c+8g+9k & 7d+8h+9l \end{bmatrix}_{3 \times 4}$$

Matrix multiplication is only defined when we have the same number of columns in A as there are rows in B.

$$\begin{bmatrix} a_{11} & a_{1n} \\ \vdots & \vdots \\ a_{m1} & a_{mn} \end{bmatrix}_{m \times n} A \quad \begin{bmatrix} b_{11} & b_{1p} \\ b_{13} & b_{1p} \\ \vdots & \vdots \\ b_{n1} & b_{np} \end{bmatrix}_{n \times p} B$$

Matrix multiplication (cont'd)

Another way to look at matrix multiplication, is to look at the columns.

Suppose we have:

$$\begin{array}{c} \vec{a}_1 \quad \vec{a}_2 \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix} = \begin{bmatrix} 1a+2b & 1c+2d & 1e+2f \\ 3a+4b & 3c+4d & 3e+4f \\ 5a+6b & 5c+6d & 5e+6f \end{bmatrix} \end{array}$$

$\underbrace{\hspace{1.5cm}}_{\vec{c}_1}$
 $\underbrace{\hspace{1.5cm}}_{\vec{c}_2}$
 $\underbrace{\hspace{1.5cm}}_{\vec{c}_3}$

Looking at the three columns of C , we identify them as linear combinations of the columns of A .

$$\begin{array}{cc} A & B \\ 3 \times 2 & 2 \times 3 \end{array} \qquad \begin{array}{c} C \\ 3 \times 3 \end{array}$$

Let's look at the second column of C , $\vec{c}_2$, in detail:

$$\vec{c}_2 = \begin{bmatrix} 1c+2d \\ 3c+4d \\ 5c+6d \end{bmatrix} = c \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + d \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = c\vec{a}_1 + d\vec{a}_2$$

Denoting the columns as vectors, we see that $\vec{c}_2$ is a linear combination of $\vec{a}_1$ and $\vec{a}_2$.

Generalized on column-vector form, we write:

$$\vec{c}_j = b_{j1}\vec{a}_1 + b_{j2}\vec{a}_2 + \dots + b_{jn}\vec{a}_n$$

Matrix multiplication (cont'd)

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Let's shift focus 90° and look at it as rows, rather than columns.

$$\begin{matrix} A \\ 3 \times 2 \end{matrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \cdot \begin{matrix} B \\ 2 \times 3 \end{matrix} \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix} = \begin{matrix} C \\ 3 \times 3 \end{matrix} \begin{bmatrix} 1a+2b & 1c+2d & 1e+2f \\ 3a+4b & 3c+4d & 3e+4f \\ 5a+6b & 5c+6d & 5e+6f \end{bmatrix}$$

Looking at C we see that the rows of C are linear combinations of rows of B.
Let's inspect the third row in detail:

$$\begin{aligned} r_3(C) &= 5a + 6b \quad 5c + 6d \quad 5e + 6f && \text{we can recognise this as being linear combination of rows of B.} \\ &= 5[a \ c \ e] + 6[b \ d \ f] \end{aligned}$$

Generalized we would write:

$$r_i(C) = a_{i1} r_1(B) + a_{i2} r_2(B) + \dots + a_{in} r_n(B)$$

Matrix multiplication (cont'd)

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A row times a column will give us a number

$$\begin{matrix} 1 \times 2 & 2 \times 1 & 1 \times 1 \end{matrix} \quad \begin{bmatrix} a & b \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} a+2b \end{bmatrix}$$

How about a column times a row?

$$\begin{matrix} 3 \times 1 & 1 \times 2 \end{matrix} \quad \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 7 \end{bmatrix}$$

Using what we know, multiplying a 3×1 by a 1×2 should produce a matrix with dimensions 3×2 .

Let's use our newly achieved skill, row multiplication and see:

The first row of C is $2 \cdot \begin{bmatrix} 1 & 7 \end{bmatrix}$

The second row is $3 \cdot \begin{bmatrix} 1 & 7 \end{bmatrix}$

And the last $4 \cdot \begin{bmatrix} 1 & 7 \end{bmatrix}$

$$= \begin{bmatrix} 2 & 14 \\ 3 & 21 \\ 4 & 28 \end{bmatrix}$$

And indeed we have a 3×2 matrix

Matrix multiplication (cont'd)

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Another way to look at matrix multiplication is as the sum of columns of A times rows of B

AB = sum of (columns of A) $\cdot$ (rows of B)

$$A = \begin{bmatrix} 1 & 7 \\ 2 & 8 \\ 3 & 9 \end{bmatrix}_{3 \times 2}, B = \begin{bmatrix} 1 & 6 \\ 2 & 7 \end{bmatrix}_{2 \times 2}$$

$$AB = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 6 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 2 & 12 \\ 3 & 18 \end{bmatrix} + \begin{bmatrix} 14 & 49 \\ 16 & 56 \\ 18 & 63 \end{bmatrix} = \begin{bmatrix} 15 & 55 \\ 18 & 68 \\ 21 & 81 \end{bmatrix}$$